

After completing the preparation materials, students should be able to:

1. **PK properties:** Explain how the main pharmacokinetic features of the antiepileptic drugs and their metabolic- and transport-related drug interactions relate to patient care.
2. **Mechanisms:** Describe the various mechanisms of antiseizure drug action at the levels of specific ion channels and neurotransmitter systems.
3. **Therapeutic uses:** Differentiate the mechanisms and effects of drugs used in the treatment of focal-onset seizures, generalized-onset seizures, various epileptic syndromes, and status epilepticus by giving specific examples.
4. **Adverse effects:** Correlate the adverse effects of the drugs covered in this lecture (the “starred drugs”) with their pharmacokinetic properties and mechanisms of action.
5. **Pharmacogenomics:** Explain the meaning and clinical relevance of HLA genetic polymorphisms on adverse event risk.
6. **Drug-drug, drug-disease, drug-age interactions:** Apply the distinctive class and drug-specific toxicities to the selection of antiepileptic therapy for the individual patients, including children, adults, pregnant patients, and older patients.

Core principles of antiseizure drug therapy

- Epilepsy is a chronic disorder of brain function characterized by the recurrent and unpredictable occurrence of seizures. Approximately 1% of the world's population has epilepsy, which is the fourth most common neurologic disorder after migraine, stroke, and Alzheimer disease.
- The antiseizure drugs (ASDs) described in this lecture are typically used chronically to prevent the occurrence of seizures in people with epilepsy.
- They are also used occasionally in people who do not have epilepsy to prevent seizures that may occur as part of an acute illness such as meningitis or in the early period following either neurosurgery or traumatic brain injury.
- Certain ASDs terminate ongoing seizures such as in status epilepticus or prolonged febrile seizures or following exposure to seizure-inducing nerve toxins.
- For seizures caused by an acute underlying toxic or metabolic disorder, such as hypocalcemia, appropriate therapy should be directed toward correcting the specific abnormality.

Core principles of antiseizure drug therapy

- **Adequate drug exposure must be maintained continuously to prevent seizures.**
- ASDs have different pharmacological profiles.
- Many ASDs have narrow therapeutic windows. Dosing must avoid toxic exposure.
- Many ASDs are metabolized in the liver. Drug interactions complicate therapy.
- ASDs are typically grouped by their principal mechanism of action. However, for many drugs, the precise mechanism is not known, or several mechanisms are supposed, based on seizure protection in diverse studies in animal models.
- Narrow spectrum agents only treat focal onset and focal evolving to bilateral tonic-clonic seizures.
- Broad spectrum agents treat a broad range of seizure types, both focal onset and primary generalized onset seizures.
- When considering discontinuation of antiseizure therapy, the risk of seizure recurrence should be weighed against the risk of side effects with continued ASD treatment. Certain seizure types and benign epileptic syndromes may remit.

Classification of Seizure Disorders

- **Focal onset seizures** arise from excessive, high-frequency, abnormally synchronous neuronal activity within a specific, discrete local population of neurons in one cerebral hemisphere. Neuronal hyperexcitability and hypersynchrony due to localized, rapidly firing voltage gated sodium channels is a hallmark of focal onset seizures. Focal seizures can also evolve into bilateral tonic-clonic seizures.
- **Primary generalized onset seizures** arise in both hemispheres simultaneously, often involving deep brain structures (thalamus, brainstem) and widespread cortical networks. The balance between excitation and inhibition across both sides of the brain is disrupted.
 - Absence seizures
 - Generalized tonic-clonic seizures
 - Infantile spasms (West syndrome)
 - Lennox-Gestaut syndrome (developmental and epileptic encephalopathy)
 - Juvenile myoclonic epilepsy
 - Dravet syndrome
- **Status epilepticus** is operationally defined as ≥ 5 minutes of continuous seizure activity or ≥ 2 discrete seizures between which there is incomplete recovery of consciousness.

Pharmacological Treatment of Seizure Disorders

Goals of therapy:

- Controlling seizures
- Avoiding treatment side effects
- Maintaining or restoring quality of life

Individualize ASD therapy:

- Treat underlying conditions
- Avoid precipitating factors
- Preventive therapy for suppression of recurrent seizures
- Address psychological and social issues

ASD therapy selection

Consider the following:

- Seizure classification and accurate diagnosis
 - Potential adverse effects of the drug
 - Comorbid medical conditions, especially but not limited to hepatic and renal disease
 - Interactions with other medications
 - Age and sex, including childbearing plans
 - Lifestyle and patient preferences
 - Cost
-
- Adequate drug exposure must be maintained continuously to prevent seizures.
 - Most ASDs follow first-order kinetics.

What is the common thread that connects this diverse group of antiseizure drugs?

ASDs protect against seizures by interacting with one or more molecular targets in the brain to inhibit the local generation of seizure discharges by:

- reducing the ability of the neurons to fire action potentials at a high rate,
- reducing neuronal synchronization,
- inhibiting the spread of epileptic activity to nearby and distant sites by:
 - strengthening the GABAergic inhibitory activity, and/or
 - inhibiting glutamate-mediated excitatory transmission.
 - stabilizing membrane potential via K⁺ channels

The 3 major categories are:

VG Na⁺ channels:

Drugs that preferentially bind to the open and inactivated state of voltage-activated Na⁺ channels on presynaptic membranes → limit the sustained, repetitive firing of neurons

VG T-type Ca²⁺ channel:

Inhibition of voltage-activated Ca²⁺ channels responsible for T-type Ca²⁺ currents in thalamocortical neurons for management of absence seizure

Modulation of various other Ca²⁺ channel types

GABA / Glutamate:

Drugs that enhance GABA-mediated synaptic inhibitory effect, and/or diminish glutamate's excitatory effects

Na⁺ and Ca²⁺ channel modulation are the largest mechanistic classes of antiseizure drugs.

Class Pharmacokinetics and Relevance To Patient Care

PK Property		The TAKEAWAYS
A	Oral IV Other routes	Oral use $\Rightarrow$ convenience for patients and caregivers IV $\Rightarrow$ rapid onset; hospital use IM, intraosseous, intranasal, rectal administration if IV access not obtained
D	Many ASDs >90% protein bound Widely distributed	Displacement can cause a temporary rise in free fraction $\Rightarrow$ transient acute toxicity (Remember: narrow therapeutic windows.) Readily diffuse across BBB to sites of action – and cause central and systemic side effects.
M	Hepatic metabolism Active metabolites	High potential for drug-drug interactions by enzyme induction or inhibition. - The ASDs' metabolism / bioavailability may be altered. - The conversion to active metabolites may be affected by other drugs. - Some ASDs are enzyme inducers or inhibitors altering the metabolism of other drugs.
F	80-100%	Good bioavailability $\Rightarrow$ predictable blood levels
E	Hepatic clearance and/or Renal clearance	Hepatic and/or renal function can affect plasma drug levels. (Remember: narrow therapeutic windows.)
$t_{1/2}$	Drug-specific	Some ASDs have short half-lives $\Rightarrow$ frequent daily administration Some ASDs have longer half-lives or come in extended-release versions $\Rightarrow$ less frequent dosing may improve adherence

Pharmacogenomics: Clinically important polymorphisms in HLA types

HLA-B*15:02: Carbamazepine / oxcarbazepine, phenytoin, lamotrigine association with Stevens-Johnson syndrome / toxic epidermal necrolysis (SJS/TEN)

HLA-B*15:02 allele occurs almost exclusively in patients of Asian ancestry. Screening is recommended for at-risk, treatment-naïve patients.

Severe cutaneous adverse reactions (SCAR) – Stevens-Johnson Syndrome (SJS) and toxic epidermal necrolysis (TEN): Severe, acute, life-threatening mucocutaneous hypersensitivity reactions can occur 1 to 4 weeks after starting the drug. Severe skin barrier loss leads to fluid loss, electrolyte imbalance, and hypothermia. Physical and psychological sequelae may occur during the chronic phase of convalescence and recovery. Management includes screening and treatment to maintain quality of life.

- SJS: Skin detachment affecting <10% of body surface area (BSA); mortality ~10%
- TEN: Skin detachment affecting >30% of BSA; mortality >30%
- SJS/TEN overlap: Skin detachment affecting 10%-30% of BSA

Sepsis, acute respiratory distress syndrome, and multiorgan failure are the most common causes of in-hospital death.

HLA-A*31:01: Carbamazepine / Oxcarbazepine strong association with **d**rug **r**eaction with **e**osinophilia and **s**ystemic **s**ymptoms (DRESS). Screening is recommended in carbamazepine-naïve patients.

HLA-A*31:01 allele is found in many populations and is more common overall: European, Japanese, Koreans, Chinese, South Asians

DRESS is a severe hypersensitivity reaction: Skin rash, visceral organ involvement, lymphadenopathy, eosinophilia, and atypical lymphocytosis. The disease course is typically prolonged and flares can occur despite cessation of the offending drug. DRESS can occur 2 to 8 weeks after initiation of the drug. Estimated mortality rate is 2%-10%. DRESS survivors should undergo long-term monitoring for the development of autoimmune sequelae¹⁵



Cautions

- **Toxicity potential:** The therapeutic index for most AEDs is low, and toxicity is not uncommon.
- **Withdrawal:** Gradually taper to minimize the potential of increased seizure frequency.
- **Suicidal ideations:** A consequence of the drug therapy? or, of the seizure disorder? (further research is needed)
- **Pregnancy:** Increased risk of congenital malformations; folic acid and vitamin K supplementation recommended
- **Pediatrics:** Larger V_d in neonates, infants, children; neonate has immature metabolic and elimination capabilities
- **Geriatrics:** Comorbid conditions; physiologic changes
- **Drug interactions:** Enzyme inducers/inhibitors; CNS depressants

Phenytoin Pharmacokinetics: Absorption and Distribution

- A** **Oral:** sodium salt or phenytoin base (“free acid” form)
100 mg phenytoin sodium $\approx$ 92 mg of free acid $\Rightarrow$ *adjust dose when switching dosage forms*
I.V.: propylene glycol (40%) and alcohol (10%) and adjusted to pH $\sim$ 12 for solubility
Avoid I.M. route: precipitation in tissues $\rightarrow$ irritation and painful local reactions, plus slow erratic absorption
➤ **Fosphenytoin:** soluble prodrug $\rightarrow$ **safer** I.M. and I.V. use

Oral absorption of phenytoin base is formulation-dependent (chew tabs; suspension) owing to variations in particle size and additives between drug formulations and manufacturer: 20% to 90%. The sodium salt (oral capsules) is nearly completely absorbed.

- D** Phenytoin distributes to brain, liver, muscle, and fat
Plasma protein binding $\sim$ 90%

Conditions that $\uparrow$ free phenytoin levels can increase toxicity risk:

- Low serum albumin levels: hypoalbuminemia (liver failure); nephrotic syndrome (protein losing); burns (capillary leakage); pregnancy (larger blood volume); age (neonates/elderly), other conditions
- Displacement from albumin sites: uremia (renal failure); hyperbilirubinemia / severe jaundice, drug interactions

Phenytoin Pharmacokinetics: Metabolism, Excretion, $t_{1/2}$

M Hepatic CYP2C9 (80%-90%), CYP2C19 (~10%-20%)
Good bioavailability (low hepatic extraction ratio)

Takeaway:

- CYP2C9 inhibitors / CYP2C9 poor metabolizers → reduced metabolism, ↑ toxicity risk
- Phenytoin is a **strong inducer** of CYP2C9, 2C19, CYP3A4, P-gp, UGT1A1
- Add CYP2C19 inhibitors / poor metabolizers = further reduce metabolism, ↑ toxicity risk.

⇒ Monitor free phenytoin levels closely. Dosage adjustments may be necessary.

E Glucuronide metabolites; minimal active drug excreted

Renal / hepatic insufficiency: Increased free fraction may occur in patients with altered kidney or liver function.

⇒ Monitoring of free (unbound) concentrations is recommended.

$t_{1/2}$ **Zero order kinetics** ⇒ **$t_{1/2}$ is dependent on plasma concentration**

$t_{1/2}$ increases with increasing drug concentration
(the V_{max} in infants >6 months and children is greater than in adults)

Recap:

- Phenytoin (base) has erratic absorption.
- Phenytoin IV solution contains propylene glycol, ethanol, and adjusted to pH ~12.
- Widely distributed; high protein binding.
- Complex metabolism and enzyme induction.
- Unpredictable levels and half-life because of zero order kinetics

PHENYTOIN exhibits Zero-Order Kinetics at therapeutic levels.

Zero-order kinetics:

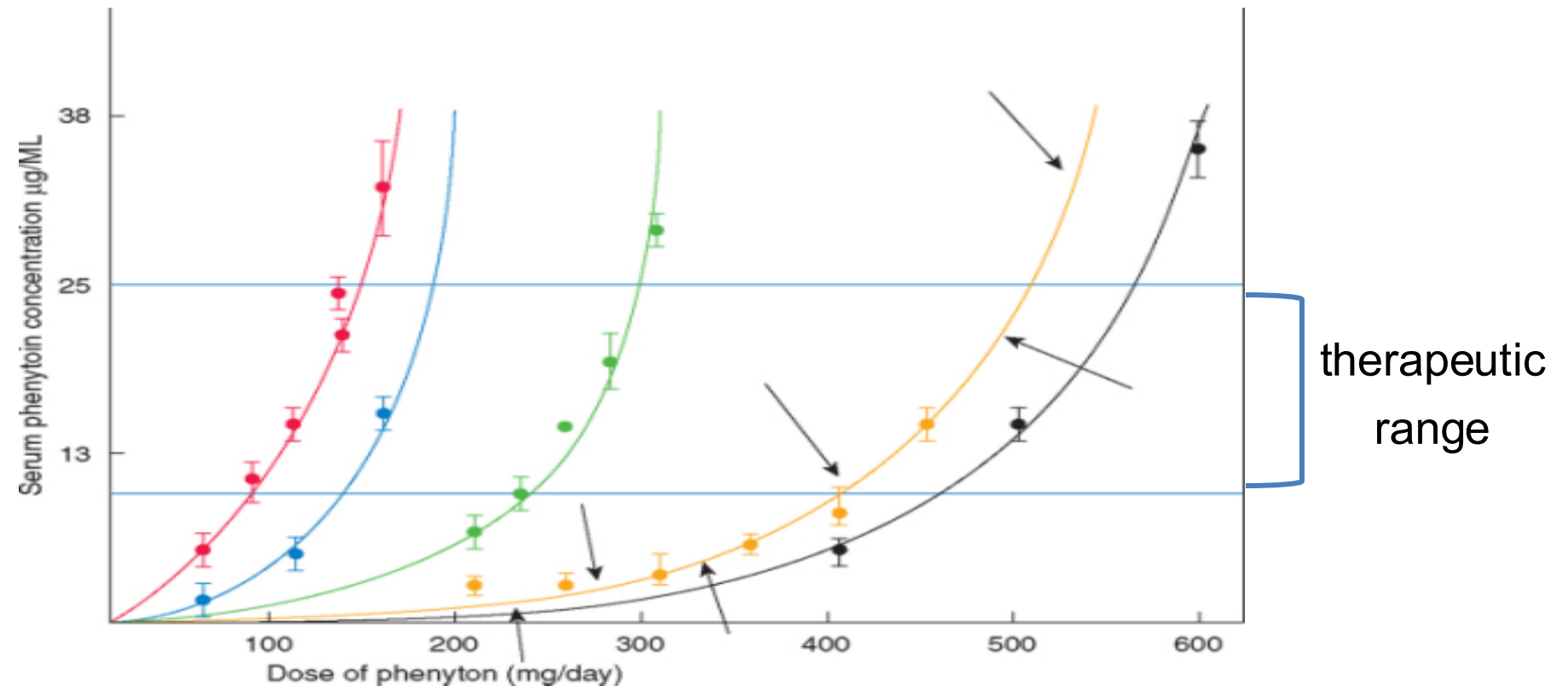
- Constant elimination of a fixed amount of drug per unit of time.

Nonlinear kinetics:

- An increase in dose produces a disproportionately large increase in serum level when metabolic and binding capacity are saturated.

Drug serum levels are **unpredictable** when changing the dose.

Clearance and elimination $t_{1/2}$ depend on the plasma concentration.



The figure shows the relationship between serum phenytoin concentration and daily dose in five patients.

Each point represents the mean ($\pm$ SD) of 3 to 8 separate measurements of phenytoin at steady state.

Saturation points differ between patients.

Clinical correlates related to phenytoin pharmacokinetics

1. **Loading doses**, I.V. or oral, are used to rapidly achieve therapeutic concentrations; monitor (check blood levels within hours)
2. **Cautious adjustment** of subsequent doses is recommended; several weeks may be required to reliably assess the results of dose changes because the time to reach steady state increases as the dose increases (nonlinear kinetics)
3. **Maintenance doses**, but not loading doses, should be lower in patients with liver failure, which causes reduced metabolism and hypoalbuminemia.

Patients with organ impairment may require lower doses:

4. **Liver failure patients:** Hypoalbuminemia and reduced drug metabolism may increase free fraction.
5. **Kidney failure patients:**
 - Nephrotic syndrome can cause hypoalbuminemia as protein is lost in the urine.
 - Kidney failure causes uremia, which may ↓ protein binding and ↑ free phenytoin concentration. (uremia: toxic accumulation of urea/nitrogenous compounds)

➤ ***Practice point: Measurement of free phenytoin may help guide dosing.***

Phenytoin Therapeutic Uses

For control of:

- Focal onset seizures (especially impaired awareness, temporal lobe)
- Evolving into generalized tonic-clonic seizures
- Status epilepticus (slow IV infusion with continuous cardiac monitoring*)
- Prevention and treatment of seizures occurring during or following neurosurgery

Efficacious, but toxicities and complex pharmacokinetics limit its use.

Historical use:

*Phenytoin is an ***anti***arrhythmic agent (like lidocaine), formerly an alternative in the treatment of ventricular arrhythmias caused by digitalis intoxication.

It has largely been replaced by newer more predictable drugs.

Major side effects	Neurotoxic effects, dose-dependent: signs of toxicity	Cardiovascular with rapid IV admin*
Hypersensitivity features: Rash, SJS, TEN (HLA-B*15:02) fever, hepatotoxicity agranulocytosis (rare)	Nystagmus / Double vision	Hypotension
	Ataxia	Bradycardia
	Slurred speech	Heart block
Folic acid deficiency (↓ absorption, ↑catabolism) Megaloblastic anemia	Sedation at high drug levels	Ventricular tachycardia
	Neuropathy – >5yr therapy or elevated drug concentrations Distal sensory loss Lower extremity areflexia Rarely, other serious polyneuropathies	Ventricular fibrillation
Vitamin D deficiency – impaired bone mineralization (Induction of hepatic Vit D catabolism) Rickets / Osteomalacia		*CV side effects are rate- and dose-dependent. Phenytoin is a Na+ channel blocker with cardiac effects. However, side effects are largely attributable to the propylene glycol solvent, which is a known vasodilator and cardiac depressant. Fosphenytoin can cause CV effects but is safer than phenytoin IV at recommended rates of administration.
Cosmetic side effects: · Gingival hyperplasia · Coarsening facial features · Hirsutism (body hair)	Infusion-related side effects: · Extravasation · Purple glove syndrome	
Drug interactions: High potential! Levels of phenytoin and levels of other drugs can be affected. Additive adverse effects ADME mechs. especially through CYP induction / displacement from binding sites.		

Phenytoin in Pregnancy

Fetal hydantoin syndrome:

Estimated 2.4-fold increased risk for any major malformation in children with prenatal exposure compared to controls

- meta-analysis data from published observational studies and pregnancy registries.

Specific fetal defects associated with chronic phenytoin use in the pregnant woman include:

- intrauterine growth retardation
- abnormalities of the skull and facial features
- cleft palate
- microcephaly
- mental retardation
- underdeveloped nails of the fingers and toes.

➤ **Folic acid throughout pregnancy is recommended.**

Breastfeeding: Infant would be exposed to an estimated 0.5%-8% of maternal weight-adjusted dosage of drug in breastmilk. Some authorities advise against breastfeeding while taking the drug.



Figure Facial features and limb abnormalities of fetal hydantoin syndrome. Clinical photographs of the patient showing A, hypertelorism, flat and broad nasal bridge; B, short upturned nose, long philtrum, thin upper lip; C, stiff tapered fingers with hypoplastic fingernails; and D, hypoplastic toenails and digitalized great toe.

Fetal Hydantoin Syndrome. Hegde, Avani et al.
The Journal of Pediatrics, Volume 188, 304; Sept 27, 2017

Mnemonic for Phenytoin Side Effects

- P** P-450, P-glycoprotein interactions (an enzyme inducer)
- H** Hirsutism
- E** Enlarged gums (gingival hyperplasia)
- N** Nystagmus
- Y** Yellowing of skin (jaundice, indicative of rare but serious hepatotoxicity)
- T** Teratogenicity
- O** Osteomalacia (impaired bone mineralization / bone softening in children, adults) – CYP induction leads to accelerated hepatic degradation of vitamin D, impaired Ca^{2+} absorption → bone resorption
- I** Interference with folic acid absorption and ↑folate breakdown → megaloblastic anemia
- N** Neuropathies: vertigo, ataxia, headache
- S** Skin coarsening; skin reactions; purple glove syndrome (I.V.)
- CV** Rapid IV administration or high dose:
Phenytoin is a Na^+ channel blocker and propylene glycol is a vasodilator and cardiac depressant
- Conduction abnormalities: Bradycardia, AV block, QRS widening, ventricular arrhythmias
 - Vasodilation: Hypotension

Check your knowledge: Fill in the column on the right.

PK Property	TAKEAWAY: Relevance to Patient Care
Oral formulations having slow erratic absorption	
Phenytoin IV Fosphenytoin IV, IM	
Plasma protein binding ~90%	
Metabolism	
Clearance is not constant	
$t_{1/2\text{elimination}}$ is highly variable.	

P
H
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CV

TAKEAWAYS: Carbamazepine is a strong CYP3A4 inducer and induces its own metabolism.

The congeners have more favorable PK profiles.

	Carbamazepine	Oxcarbazepine	Eslicarbazepine acetate
A D	Oral Mid-high protein binding	Oral prodrug Moderate protein binding	Oral prodrug Low-moderate protein binding
M	CYP3A4 metabolism Bioavailability 75%-85%	Rapid reduction to active MHD = S-licarbazepine (predominantly) P450 enzymes not involved. Good MHD bioavailability	Rapid hydrolysis to active S-licarbazepine (major) and R- licarbazepine (minor) Bioavailability >90%
E	Urine/feces as metabolites negligible unchanged drug	Urine: MHD (27%), glucuronide and other metabolites	Urine: Licarbazepine (66%), glucuronide and other metabolites
t _{1/2}	Variable, 25-65 hours because of autoinduction, which is usually complete in 3-5 weeks	Oxcarbazepine: ~2h MHD: ~9 hours	S-licarbazepine 13-20 hours
	CYP3A4 Strong inducer → <i>Induces own metabolism (autoinduction)</i> Also induces, 2B6, 2C9, 1A2, UGT1A1, P-gp, MRP2, OATs	CYP3A4 Weak inducer Does not induce own metabolism	CYP3A4 moderate inducer Does not induce own metabolism

Carbamazepine and congeners: Therapeutic Uses

Carbamazepine:

Gold standard – First-line option for treatment of focal epilepsy in all ages

CBZ is highly effective – It has been a comparator in nearly all modern trials and it is widely available, affordable, and long clinical experience

- Focal onset seizures and
- Focal evolving to generalized tonic-clonic seizures
- *Not effective and may increase frequency in absence, myoclonic, and akinetic seizures.*

Nonseizure applications

- Trigeminal neuralgia
- Hypomania and mild to moderate mania associated with bipolar disorder
- Bipolar major depression

Oxcarbazepine and Eslicarbazepine

- Monotherapy or adjunctive treatment of focal-onset seizures in children and adults

Nonseizure applications:

Oxcarbazepine:

- Trigeminal neuralgia
- Acute mania associated with bipolar disorder – adjunct when first-line agents are ineffective or poorly tolerated

Carbamazepine dosing due to autoinduction:

The carbamazepine is increased gradually based on seizure control, tolerability, and serum concentrations.

Adverse effects and cautions: CBZ, OXC, and ESL

Acute, dose-related	dizziness, sedation at high dose, ataxia, nystagmus, nausea
Hypersensitivity reactions Rash: 3-20 days SCARS: 1-12 weeks after start of therapy	Severe cutaneous adverse reactions (SCARs), T cell-mediated: SJS/TEN; DRESS (including hepatotoxicity) HLA-B*15:02 phenotype (Asian populations) and HLA-A* 31:01 phenotype (European, Japanese, Korean, other Asian populations) Genotype screening prior to initiation of therapy in at-risk populations
Hyponatremia	Decreased serum sodium due to nephrogenic syndrome of inappropriate antidiuretic hormone (SIADH): Promotes insertion of aquaporins (AQP-2) in distal nephron → unregulated “free” water reabsorption → hyponatremia and water intoxication can occur OXC greatest risk (incidence ~10%-73%) > CBZ (incidence ~1%-40%) ≥ ESL
Hematologic effects (low incidence, reversible)	Both bone marrow suppressive (↓ synthesis of red cells, white cells, and platelets) and proliferative effects (↑ synthesis of blood cells). Mechanism not known.
Pregnancy	CBZ: Increased incidence of spina bifida, craniofacial defects, cardiovascular malformations. Other agents should be considered in patients of childbearing potential. OXC: may be considered during pregnancy; ESL: insufficient data
Drug interactions	High risk, numerous – CNS effects and metabolism/transport-related effects

SJS: Stevens-Johnson syndrome; TEN: Toxic epidermal necrolysis; DRESS: Drug reaction with eosinophilia and systemic symptoms – high fever, rash, hematologic and internal organ involvement

Check your knowledge

What are the meaning and effect of “fast inactivation enhancers”?

What are the meaning, mechanism, and effect of carbamazepine autoinduction?

Why is it important to screen at-risk populations for HLA-B*15:02 and HLA-A*31:01 variant alleles before starting carbamazepine therapy?

Check your knowledge

What are the clinical applications of carbamazepine?

Carbamazepine, oxcarbazepine, and eslicarbazepine should be avoided in what seizure types? Why?

What is the mechanism of carbamazepine-induced hyponatremia?

What effects in the newborn have been associated with carbamazepine use during pregnancy?

What are the metabolism- and transport-related drug interactions associated with carbamazepine use?

Lamotrigine Pharmacology

Pharmacokinetic profile

A	Oral (various dosage forms)
D	Protein binding 55%
M	Hepatic glucuronidation Bioavailability 98% (immediate release) (without concomitant inducers or inhibitors)
E	Urine, glucuronide metabolites
t _{1/2}	~24 h • t_{1/2} more than doubled in patients taking valproate • t_{1/2} reduced in patients taking strong broad-spectrum inducer (carbamazepine, phenytoin, phenobarbital, primidone) • t_{1/2} is increased in liver insufficiency

Dosing guidelines and color-coded starter kits are available to guide dosing and help patient adherence when initiating therapy with valproate or a metabolism-inducing ASD.

Therapeutic uses:

Lamotrigine's broader spectrum of seizures suggests that lamotrigine may have actions in addition to Na⁺ channel inhibition.

Broad spectrum

- Focal onset seizures
- Primary generalized tonic-clonic seizures
- Lennox-Gestaut syndrome
- Absence epilepsy (second-line)

Nonseizure uses

- Bipolar disorder
maintenance treatment – monotherapy or add-on
For preventing major depressive episodes
Lamotrigine does not reliably prevent manic episodes.

Lamotrigine adverse effects and cautions

Most common:

- Nausea, dizziness, drowsiness
- Low incidence of sedation and cognitive side effects compared with other ASDs
- Can worsen migraine headaches

• Serious hypersensitivity reactions:

SJS /TEN / DRESS

- usually within 2-3 months
- Screen for *HLA-B*15:02* variant allele before starting therapy in at-risk patients of Asian descent

To reduce the risk of rash, the drug should be **slowly titrated over 6 to 8 weeks** – even more cautiously if added to valproic acid.

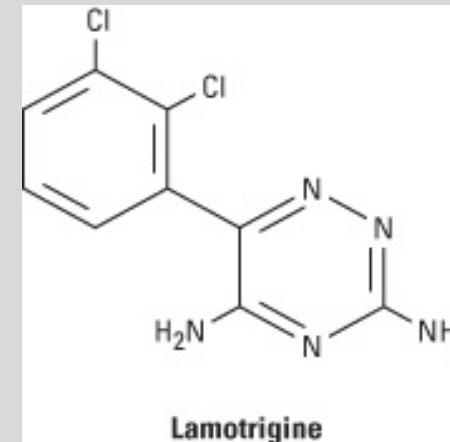
PREGNANCY Lamotrigine may be considered in patients of childbearing potential.

Lamotrigine monotherapy has been associated with lower risks of major congenital malformations compared to some other ASDs.

DRUG INTERACTIONS

- **Phenytoin, carbamazepine, and phenobarbital** induce metabolism of lamotrigine (~50% decrease in $t_{1/2}$)
- **Valproic acid** inhibits lamotrigine's metabolism (~2-3x increase in $t_{1/2}$)

KNOW



*Lacosamide

Enhances slow inactivation state

Slow inactivation state: Lacosamide selectively enhances the slow inactivation state, which stabilizes the Na⁺ channel in a long-lasting (seconds) nonconductive state. (Predominant mechanism of action.)

Fast inactivation state: Lacosamide also enhances the fast inactivation state, but with slower binding/unbinding kinetics.

(This mechanism is suggested in more recent studies.)

Effect: Lacosamide preferentially targets pathologically overactive neurons, preventing repetitive firing during sustained bursts, without broadly suppressing normal brain function (physiological fast inactivation) → better seizure control with fewer generalized side effects.

Lacosamide pharmacology

PK	Oral, IV; minimal protein binding; hepatic metabolism (minor CYP); inactive metabolites; 100% oral bioavailability; excretion urine 40% unchanged; $t_{1/2}$ ~13 h
Uses	• Focal onset and primary generalized onset seizures • monotherapy and add-on; adults and children • Off-label: status epilepticus • Lacosamide is a well tolerated alternative to carbamazepine
Adverse effects	Generally well tolerated • Common: dizziness, drowsiness, headache, ataxia, confusion, others • PR interval increase (bradycardia, AV block); (ECG recommended prior to start) ○ Risk factors: cardiovascular disease, cardiac conduction anomalies • Hypersensitivity reactions reported ranging from mild skin rash → SJS/TEN and DRESS (no large-scale evidence of association with HLA-B*15:02 variant allele) • Suicidal ideations and tendencies have been reported in association with lacosamide.
Drug interactions	CNS depressants may increase CNS depressant effects

Check your knowledge

What are the mechanisms of action of lamotrigine and lacosamide?

What is the clinical significance of the lamotrigine and lacosamide pharmacology compared with that of the other sodium channel blockers?

Check your knowledge

Why does lamotrigine have specific dosing requirements when used as an add-on agent with valproate or phenobarbital?

What are the lamotrigine dosing adjustments (in general)? Think about based on what you know.

What is the pathophysiologic basis of lamotrigine-associate skin reactions?

Check your knowledge

Risk factors for a certain potential adverse effect of lacosamide are cardiovascular disease and conduction anomalies. What is the adverse effect?

Although lacosamide has no evidence of an association with HLA-A or HLA-B variant alleles, what adverse effects have been reported with the use of this ASD?

What are the clinical applications of lamotrigine and lacosamide?

- Effective antiseizure drugs (ASDs) interact with one or more molecular targets in the brain. Based on in vitro studies, ASDs inhibit seizure generation by various mechanisms. Some ASDs reduce the ability of neurons to fire high frequency action potentials. Others strengthen inhibitory effects of GABAergic interneurons and/or reduce glutamate-mediated excitatory neurotransmission. Some ASDs have multiple mechanisms.
- Diagnosis of a specific seizure type is important for prescribing the most appropriate ASD therapeutic regimen. Drug choice is usually based on established efficacy in the specific seizure state, the prior responsiveness of the patient, and the anticipated toxicity of the drug.
- Treatment may involve combinations of drugs, following the principle of adding known effective agents if the preceding drugs are not adequate in controlling seizure activity. Therapeutic monitoring of plasma levels and clearance in individual patients may be necessary in guiding dosing for optimal therapy in the individual patient.
- Chronic therapy with ASDs is associated with specific toxic effects. Consideration of their pharmacokinetic properties and adverse effect profiles are important for optimizing therapy and avoiding toxicity and drug interactions.
- In general, antiseizure drugs are well absorbed orally and have good bioavailability. Most antiseizure drugs are metabolized by hepatic enzymes, and in some cases active metabolites are formed.

Summary

- Children born of mothers taking anticonvulsant drugs have an increased risk of congenital malformations. Fetal hydantoin syndrome has been described after phenytoin use by pregnant women. Carbamazepine has been implicated as a cause of craniofacial anomalies and spina bifida. Neural tube defects (eg, spina bifida) are associated with the use of valproic acid.
- Sodium channel blockers preferentially bind a site within the pore of the channel when the channel is in the open and *inactivated* state.
- Phenytoin, carbamazepine, and lamotrigine are enhancers of sodium channel fast inactivation, meaning they stabilize and prolong the fast inactivation state, allowing the block to accumulate, meaning that before the channel fully recovers, the next depolarization arrives → more channels enter drug-bound inactivated state (the basis of use-dependent block).
- Lacosamide selectively enhances the slow inactivation state, stabilizing the Na⁺ channel in a long-lasting (seconds) nonconductive state.
- Phenytoin, carbamazepine, lamotrigine, and lacosamide are used as monotherapy or adjuncts in the management of focal onset seizures and focal seizures evolving to generalized tonic-clonic seizures.
- Lamotrigine and lacosamide are also used as adjuncts in primary generalized seizures.

Summary (Part 1): Adverse effects profiles of the sodium channel blockers

Antiepileptic Drug	Adverse Effects
Phenytoin	Nystagmus, diplopia, sedation, gingival hyperplasia, hirsutism, anemias, peripheral neuropathy, rickets/osteomalacia, SJS/TEN (HLA-B*15:02 variant) Fetal hydantoin syndrome, induction of hepatic drug metabolism
Carbamazepine	Diplopia, cognitive dysfunction, ataxia; hyponatremia, SJS/TEN/DRESS (HLA-B*15:02, HLA-A*31:01); rare occurrence of severe blood dyscrasias; fetal malformations; induces hepatic drug metabolism
Oxcarbazepine Eslicarbazepine	As for carbamazepine, but hyponatremia is more frequent with OXC (greater propensity than carbamazepine and eslicarbazepine) Induce CYP3A4 but fewer drug interactions than carbamazepine and are not autoinducers
Lamotrigine Enhances fast inactivation state	Low incidence of sedation and cognitive side effects compared with other ASDs Skin rash / SJS/TEN (HLA-B*15:02): Start low, Go slow: Slowly titrate dose over 6-8 weeks to reduce severe cutaneous adverse reactions (SCARs) Dosing guidelines for concomitant valproate or enzyme inducer
Lacosamide Enhances slow inactivation state	Generally well tolerated; PR interval increase (bradycardia, AV block); (ECG recommended prior to start) SJS/TEN can occur but have not been associated with HLA-B*15:02 variant. 54

Check your knowledge: Fill in the column on the right.

PK Property	TAKEAWAY: Relevance to Patient Care
Oral formulations having slow erratic absorption	Difficult to predict the effect of phenytoin base formulations between patients and in one patient at different times
Phenytoin IV Fosphenytoin IV, IM	Highly alkaline solution / propylene glycol, alcohol – Slow infusion into a large vein, severe local irritation, closely monitor for extravasation. Avoid IM (precipitation). Fosphenytoin preferred – water soluble, faster infusion, safer, rarely irritating; can be given IM if vein not available.
Plasma protein binding ~90%	Conditions that increase free phenytoin level increase toxicity risk: Low albumin: hypoalbuminemia, nephrotic syndrome, age (neonate / elderly) Displacement: uremia (kidney failure)
Metabolism	Hepatic CYP2C9, 2C19, 3A4 Inducer: CYP3A4, CYP2B6, CYP2C9, 2C19, UGTs, and P-glycoprotein
Clearance is not constant	Zero-order kinetics, nonlinear: A small change in dose results in a disproportionately large change in serum concentration.
$t_{1/2\text{elimination}}$ is highly variable.	Metabolism follows zero-order kinetics. Elimination half-life increases with increasing concentration. When metabolic enzymes are saturated, a fixed amount of drug is metabolized per unit of time.

Check your knowledge of adverse effects: Define the terms of the acronym

P	P 450 and P-glycoprotein interactions
H	Hirsutism
E	Enlarged gums – gingival hyperplasia
N	Nystagmus
Y	Yellowing of skin – jaundice due to rare but serious hepatotoxicity
T	Teratogenicity – Fetal hydantoin syndrome – some or all features
O	Osteomalacia – induction of hepatic Vit D breakdown → ↓ calcium uptake
I	Interference with folate absorption and induction of folate breakdown
N	Neuropathies – vertigo, ataxia, nystagmus; polyneuropathies with long-term use
S	Skin coarsening; skin reactions; purple glove syndrome (I.V.)
CV	Rapid IV administration or high dose: Phenytoin is a Na⁺ channel blocker and propylene glycol is a vasodilator and cardiac depressant • Conduction abnormalities: Bradycardia, AV block, QRS widening, ventricular arrhythmias • Vasodilation: Hypotension

Check your knowledge

What are the meaning and effect of “fast inactivation enhancers”?

- Physiology: In less than a millisecond after opening and conduction Na⁺, channels undergo a conformational change (gating) and enters the inactivated state (fast inactivation).
- ASDs that are ***fast inactivation enhancers*** bind a site within the pore of the channel when the channel is in the *open and inactivated* state, causing it to be *stabilized* in this state, accumulate block during repetitive firing, and bind more strongly at depolarized potentials. High frequency action potentials, as in seizure activity, are preferentially inhibited.
- Phenytoin, carbamazepine, oxcarbazepine, eslicarbazepine, and lamotrigine

What are the meaning, mechanism, and effect of carbamazepine autoinduction?

- Carbamazepine is metabolized by CYP3A4 and it is a strong CYP3A4 inducer.
- It induces its own metabolism – autoinduction – which stabilizes in 3 to 5 weeks of therapy.
- Therefore, the carbamazepine dose needs to be increased gradually based on seizure control, tolerability, and serum concentrations.

Why is it important to screen at-risk populations for HLA-B*15:02 and HLA-A*31:01 variant alleles before starting carbamazepine therapy?

- Patients with HLA-B*15:02 or HLA-A*31:01 phenotypes are at risk for rash and severe cutaneous adverse effects – SJS/TEN and DRESS.

Check your knowledge

What are the clinical applications of carbamazepine?

- Seizure applications: focal onset seizures and focal evolving to generalized tonic-clonic seizures.
- Nonseizure applications: trigeminal neuralgia; bipolar disorder associate hypomania, mania, and depressive episodes.

Carbamazepine, oxcarbazepine, and eslicarbazepine should be avoided in what seizure types? Why?

- CBZ, OXC, and ESL are ineffective for and may worsen absence, myoclonic, and akinetic seizures.

What is the mechanism of carbamazepine-induced hyponatremia?

- Nephrogenic syndrome of inappropriate antidiuretic hormone (SIADH): CBZ, OXC, and ESL cause upregulation of AQP-2 in the distal nephron, resulting in unregulated water reabsorption, which leads to low serum sodium levels (dilutional hyponatremia).

What effects in the newborn have been associated with carbamazepine use during pregnancy?

- CBZ is associated with Increased incidence of spina bifida, craniofacial defects, cardiovascular malformations. Other agents should be considered in patients of childbearing potential.

What are the metabolism- and transport-related drug interactions associated with carbamazepine use?

- CBZ is a strong inducer of CYP3A4 and induces, 2B6, 2C9, 1A2, UGT1A1, P-gp, MRP2, OATs₉

Check your knowledge

What are the mechanisms of action of lamotrigine and lacosamide?

- Lamotrigine: Enhances fast inactivation so recovery is slower, which leads to progressive reduction in available sodium channels and may inhibit the release of glutamate and aspartate.
- Lacosamide: Selectively enhances the slow inactivation state, stabilizing the Na⁺ channel in a long-lasting (seconds) nonconductive state (predominant mechanism) and binds to the fast inactivation state, but with slower binding/unbinding kinetics. Lacosamide preferentially prevents seizure activity without broadly suppressing normal brain function (physiological fast inactivation) → better seizure control with fewer generalized side effects.

What is the clinical significance of the lamotrigine and lacosamide pharmacology compared with that of the other sodium channel blockers?

- Compared with carbamazepine, lamotrigine is effective for focal onset, focal evolving to generalized tonic-clonic seizures, and absence seizures with a better pharmacokinetic profile. Titration to therapeutic dose is slower to reduce risk of severe skin reactions.
- Lacosamide is used for focal onset, focal evolving to generalized tonic-clonic seizures, and add-on for refractory primary generalized seizures. It has more favorable PK and adverse effect profiles and can be titrated more rapidly than lamotrigine.

Check your knowledge

Why does lamotrigine have specific dosing requirements when used as an add-on agent with valproate or phenobarbital?

- $t_{1/2}$ more than doubled in patients taking valproate (metabolic enzyme inhibition)
- $t_{1/2}$ reduced in patients taking strong broad-spectrum inducer (carbamazepine, phenytoin, phenobarbital, primidone)

What are the lamotrigine dosing adjustments (in general)? Think about based on what you know.

- Valproate: Very slow dose titration (every other day)
- Inducers: Initiate at higher doses than when not taking and inducer.

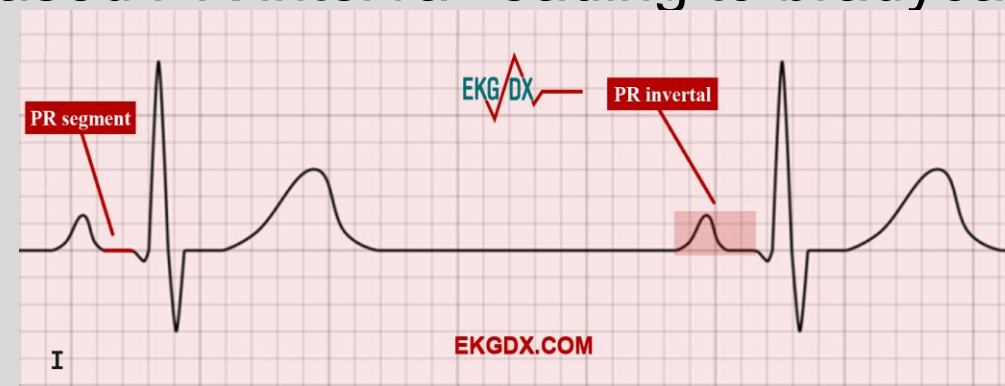
What is the pathophysiologic basis of lamotrigine-associated skin reactions?

- The likelihood of rash and severe cutaneous adverse reactions, SJS/TEN, is increased in patients with the HLA-B*15:02 phenotype.
- Screen at-risk patients before starting therapy and initiate at low doses and slowly titrate based on response and tolerability.

Check your knowledge

Risk factors for a certain potential adverse effect of lacosamide are cardiovascular disease and conduction anomalies. What is the adverse effect?

- Lacosamide can cause increased PR interval leading to bradycardia and AV block.



Although lacosamide no evidence of an association with HLA-A or HLA-B variant alleles, what adverse effects have been reported with the use of this ASD?

- Reported hypersensitivity reactions ranging from mild skin reactions to severe cutaneous reactions.

What are the clinical applications of lamotrigine and lacosamide?

- Lamotrigine: Focal onset seizures, primary generalized epilepsy, generalized seizures of Lennox-Gestaut syndrome; prevention of major depressive episodes of bipolar disorder
- Lacosamide: focal onset seizures, focal evolving to generalized tonic-clonic seizures, add-on for resistant primary generalized seizures.

Bonus: Answer to the GRQ

What are the hallmarks of so-called “favorable” pharmacokinetic profiles?

Fundamental concepts:

A drug must reach the site of action and maintain a therapeutically effective concentration at the target site for a long enough duration with minimal undesired side effects.

A drug should have a wide therapeutic window to allow some variability in dosing without major safety concerns.

A drug's effect should correlate well with its plasma concentration for easier clinical management of the dose-response relationship.

- A: High, reliable bioavailability and a rate of absorption that achieves C_{max} (peak concentration) without causing a spike in concentration that causes adverse effects.
- D: A drug should distribute to target tissues with minimal accumulation in off-target tissues (ideally) to minimize toxicity. Predictable free drug concentrations because of predictable protein binding. Ability to cross the BBB if the therapeutic effect is in the CNS; restricted ability to cross the BBB if the drug is not intended for CNS effects.
- M / E: A drug should exhibit predictable and consistent clearance with balanced renal and hepatic elimination not reliant on a single organ system, which may reduce the risk of toxic drug accumulation in organ impairment.
- t_{1/2}: A drug should have a reasonable half-life (not too short, not too long) to maintain drug levels at convenient dosing intervals but short enough to avoid excessive accumulation and prolonged toxicity if adverse effects occur
- Drug interactions: Low potential for drug interactions by having a minimal propensity to inhibit or induce metabolic enzymes and transporters.